

Lumbar disc herniation diagnosis using deep learning on MRI

Jun Qian^a, Guanghui Su^a, Xiaolin Shu^a, Kuanghong Shen^a, Bing Chen^a, Xiangdong Wang^{b,*}

^a Department of Orthopedics, Affiliated Hengyang Hospital, Hunan Normal University (Hengyang Central Hospital), Hengyang, Hunan, 421001, China

^b Hengyang Maternal and Child Health Hospital, Hengyang, Hunan, 421001, China

ARTICLE INFO

Keywords:

Artificial intelligence
Deep learning
Cervical spondylosis
Lumbar disc herniation
Image segmentation

ABSTRACT

Objective: In recent years, artificial intelligence deep learning has gradually become a research hotspot in diagnosing and treating cervical and lumbar degenerative diseases, represented by cervical spondylosis and lumbar disc herniation.

Methods: Retrospective inclusion of 590 patients with lumbar disc herniation, randomly divided into a training group (n = 456) and a testing group (n = 134), labeled with intervertebral disc herniation areas on lumbar MR images (1157 in the training group and 395 in the testing group). Construct a deep learning model based on Unet to segment and detect intervertebral disc herniation areas in sagittal and transverse planes and evaluate the model's lesion detection efficiency and accuracy in lesion segmentation.

Results: The model has good detection performance on the test group data. The F1 values in the sagittal and transverse positions are 0.971 (95% CI: 0.951–0.987) and 0.903 (95% CI: 0.870–0.935). In the segmentation of prominent areas, the Dice values in the sagittal and transverse planes are 0.774 ± 0.193 and 0.647 ± 0.261 (2D plane), 0.777 ± 0.148 and 0.655 ± 0.180 (3D plane), respectively.

Conclusion: The deep learning model based on U-net can accurately identify and segment the intervertebral disc herniation area in lumbar MR images, which helps to assist clinical diagnosis.

1. Introduction

Low back pain (LBP) is a severe public health problem worldwide, affecting 70%–80% of adults. Currently, the lifetime prevalence of LBP in the population is as high as 85% (Bhattarai et al., 2023; Yu et al., 2023). Degenerative changes in the lumbar spine are the leading underlying causes of low back pain. These include pathological and physiological changes, such as intervertebral disc degeneration, vertebral endplate degeneration, and spinal stenosis (Zhou, 2022). MRI is the most commonly used imaging technique for studying LBP. Clinical evaluation mainly relies on MRI imaging data to evaluate the signals and morphological changes of anatomical parts such as intervertebral discs, vertebral bodies, and spinal canals, assisting clinical physicians in selecting diagnosis and treatment plans. It is in the semi-quantitative evaluation stage of artificial vision and lacks accurate measurement of imaging parameters.

Moreover, the clinical diagnosis process relies heavily on the experience of physicians, and there are differences in the diagnoses and surgical plans made by different physicians (Yuan et al., 2022). However, radiologists still fail to address issues, such as high workload, low

efficiency, and poor consistency in manual measurement of lumbar anatomical parameters (Qiu et al., 2021a). Therefore, artificial intelligence (AI) technology with automatic quantitative, qualitative analysis, and diagnostic capabilities can optimize the diagnostic procedures of radiologists, improve disease diagnosis accuracy, and assist clinical physicians in diagnosing lumbar spine diseases and formulating surgical plans through lumbar MRI imaging, thereby improving medical quality (Cheung & Hong, 2023).

Presently, domestic and foreign scholars often use the three main categories of AI in LBP: computer vision technology, computer-aided diagnosis, and decision support systems. Its advantages lie in automatic feature extraction and image segmentation, recognition of pathology and quantification of disease levels, prediction of disease progression, etc. It significantly improves the work efficiency of physicians and reduces or even eliminates minor differences between observers (Qiu et al., 2021b). Machine learning (ML) is a component of AI, which is the process by which a computer automatically learns and executes predefined tasks without prior knowledge. Deep learning (DL) is a new research direction in the field of ML, which has strong self-learning ability and eliminates more human factors than traditional

* Corresponding author.

E-mail address: wxd197366@163.com (X. Wang).

<https://doi.org/10.1016/j.jrras.2024.100988>

Received 24 February 2024; Received in revised form 30 May 2024; Accepted 31 May 2024

Available online 7 June 2024

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machine learning models. For example, deep learning algorithm models such as support vector machines (SVM), artificial neural networks (ANN), and convolutional neural networks (CNN) have been widely used in the field of medical image analysis. Especially in intelligent imaging-assisted diagnosis (A et al., 2022), its application in lumbar MRI has become a research hotspot. Therefore, this paper intends to analyze the current application status and development trend of DL technology in lumbar MRI from automatic recognition of lumbar spine structure by DL, automatic diagnosis and grading of LBP-related diseases, and prediction of surgical necessity.

2. Related work

2.1. Research progress in deep learning for the diagnosis of cervical spondylosis

Due to the close correlation between the diagnosis of cervical and lumbar degenerative diseases and imaging data, many scholars have begun to use machine learning models and deep learning techniques to extract structural features of the cervical and lumbar vertebrae and assist in disease diagnosis. At present, the diagnosis of cervical spondylosis mainly relies on the evaluation of MRI images, especially analyzing the compression of intervertebral discs and surrounding nerve tissue on T2-weighted images to clarify the diagnosis of the disease (Batcik et al., 2021), which also creates good conditions for deep learning models. In recent years, research has found that deep learning models can provide precise localization of spinal cord lesions in patients with cervical spondylosis. CNN models automatically detect and label vertebral sequences on cervical MRI images, with an accuracy rate of over 99% and a sensitivity of 99.1%–99.8% (Wu et al., 2021). Wang et al. (Shi et al., 2021) used diffusion equivalent imaging (DTI) with a DTI feature extractor to extract features from cervical magnetic resonance images of a population. They analyzed them using machine learning algorithms to predict cervical spondylosis diagnosis accurately. However, this method poses special requirements for magnetic resonance imaging, as the amount of data that can be obtained is relatively tiny, and manually designed feature extractors make it challenging to meet the diagnosis of different types of cervical spondylosis. Subsequently, Jin et al. (Wang et al., 2021) further optimized the feature extraction process of artificial intelligence. They used different models to compare the learning and analysis of MRI manifestations and the prognosis of cervical spondylitis myelopathy. They found that the support vector machine (RBF-SVM) method using the Gaussian kernel function can significantly improve the ability of DTI imaging to analyze the MRI spinal cord feature. The accuracy and sensitivity of disease diagnosis and prognosis analysis based on this model reached 85.0% and 92.4%, respectively.

2.2. Prediction and qualitative diagnosis of lower back pain using deep learning models

Lower back pain is a common disease among middle-aged and older adults, mainly caused by lumbar disc herniation compressing nerves, so its onset has a certain regularity. Hu et al. (He et al., 2020) established an extended short-term memory deep learning model that can predict the occurrence of lower back pain by recording motion data of different spine segments in the population and combining it with lower back pain status. Based on this concept, the input of deep learning can be further diversified, such as using relevant data such as gender, age, height, weight, smoking history, drinking history, and weight-bearing habits as the input, and whether there is lower back pain as the output, to assist in diagnosing the occurrence of lower back pain. A study has found that through the above methods, there is a strong correlation ($r = 0.997$) between the predicted occurrence of lower back pain using CNN models and the actual symptoms of lower back pain, indicating that CNN models can effectively predict the occurrence of lower back pain through

imaging and clinical parameters, providing the possibility of early diagnosis (Gadjradj et al., 2020); Meanwhile, for the correlation between specific changes in spinal mobility and imaging parameters and different categories of spinal diseases, the use of CNN models to “validate” data produced better results compared to subjective predictions by physicians, with an accuracy rate of up to 85% (Soo, 2020). Therefore, the rational use of deep learning models can assist in diagnosing lumbar degenerative diseases and improve the work efficiency of physicians to a certain extent.

For lumbar degenerative diseases such as lumbar disc herniation, clinical diagnosis mainly relies on imaging results to determine the specific lesion and severity of the condition. Therefore, deep learning is also suitable for carrying out corresponding work. For example, Jama-ludin et al. (Mo et al., 2019) used a CNN model with lumbar magnetic resonance imaging as the input, lumbar disc Pfirrmann score (5 grades), intervertebral height (4 grades), presence of intervertebral disc herniation, presence of spinal stenosis, presence of endplate damage, and presence of vertebral body Modic changes as the output. After training the model, Its accuracy in analyzing new data output (compared to manual diagnosis by imaging experts) reaches 95.6%. Han et al. (Yu et al., 2019) used a CNN model based on multi-scale and multitasking learning networks, using T1/T2 weighted magnetic resonance imaging of the lumbar spine as the input and the position and presence of lesions of the intervertebral disc, intervertebral foramen, and vertebral body as the output. They achieved an accuracy of over 90% in diagnosing intervertebral foramen stenosis, intervertebral disc degeneration, and vertebral body lesions through imaging images. In addition, research has been conducted on how to achieve automatic recognition of intervertebral disc segments through the CCN system, identifying extraordinary imaging manifestations of lumbar degeneration such as fused vertebrae, scoliosis, lumbar sacralization, and identifying intervertebral foramen size and nerve root compression. Although the typical imaging data of the above manifestations are less than conventional degeneration, the learning of a small number of cases still demonstrates high-resolution accuracy; avoiding the subjectivity and diversity of diagnosis by different physicians is of great significance for the diagnosis, treatment, and prognosis of patients (Wu et al., 2019).

3. Method

The overall network adopts a U-net encoder + decoder structure, as shown in Fig. 1. The blue network represents the segmentation network of the MR sagittal image modality. In contrast, the orange network represents the segmentation network of the MR transverse image modality. The encoder parts of both networks use 3D resnet18 and do not share parameters. Considering the parameter redundancy of the 3D network and the high memory consumption of the transformer structure during training, the number of channels in each layer of resnet18 is modified to 16,32,64,128,192. In addition, due to the limited number of layers in the z-axis direction of lumbar spine MR sagittal and transverse images, no downsampling is performed on the z-axis in the network, and sampling operations are performed on the x-axis and y-axis. Meanwhile, based on the batch size feedback during 3D network training, all “batch normalization” operations will be replaced with “group normalization” operations, with a group size of 4.

Because sagittal and coronal MR images are data from the same patient with different modalities, there is a certain degree of complementarity in the information between the two modalities. If the feature information of the two modalities is fused, it can better segment them. Therefore, after encoding the features in the encoder, we adopted a multimodal mutual attention mechanism based on a transformer (A et al., 2022) to fuse and use the features of the two modalities, achieving better segmentation results. In the sagittal and transverse MR inputs, two corresponding features, Sagittal and Axial, can be obtained by passing through their respective encoder structures. The two are input into the multimodal mutual attention mechanism module based on the

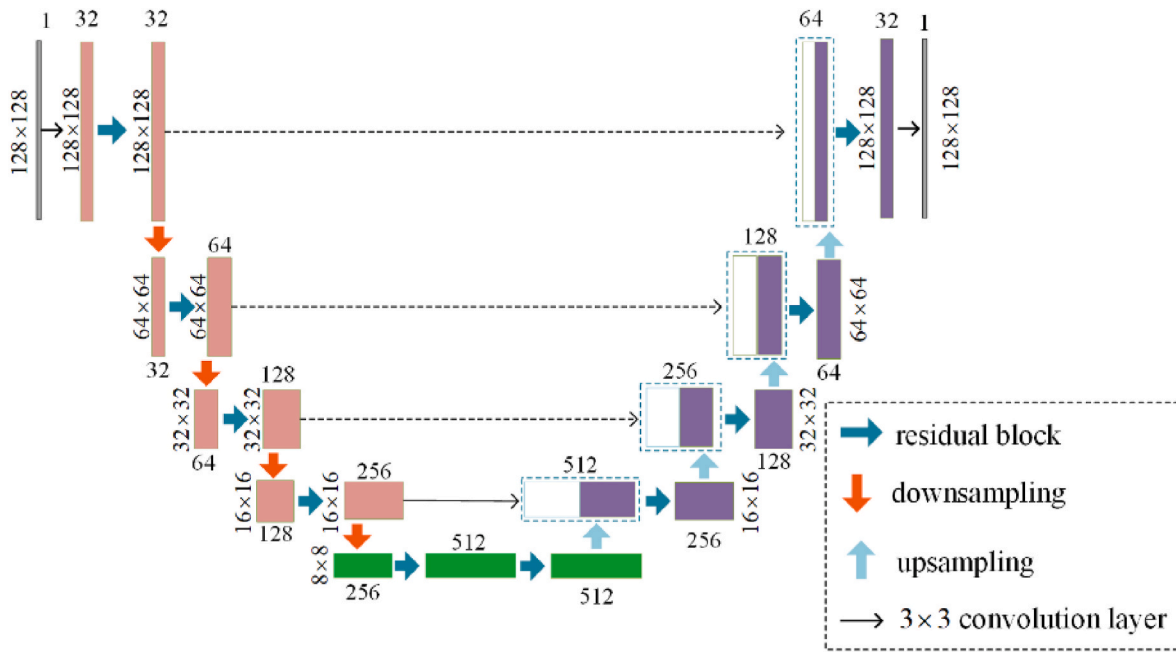


Fig. 1. Schematic diagram of U-net form encoder and decoder structure.

transformer, and feature enhancement is performed on the Sagittal and Axial image data, respectively, so that each feature can be enhanced through multimodal information. Specifically, taking Sagittal as an example, in the transformer, three feature vectors, K, Q, and V, are required for calculation. When enhancing Sagittal features, Q is calculated through Sagittal encoding, and K and V are calculated through XAxial encoding. Through the multi-head attention mechanism (Yu et al., 2023) method (with eight heads), the enhanced features of Sagittal are obtained and returned to the U-net network as the lowest-level feature, continuing the decoder process. This strengthens the interaction between the two modalities, allowing for greater utilization of multimodal information in the network. During the decoding process, the deep features are unsampled to the size of adjacent shallow features, and feature concatenation is performed. This operation is repeated continuously until the features are restored to the original input size, and the final network prediction result is obtained through the activation function. We train each modality using a mixed loss function of dice loss and focal loss (Gil et al., 2022). The network uses the Adam optimizer, with a learning rate of 0.0001 and a batch size of 4, for 300 generations.

4. Experiments and results

4.1. Dataset

A total of 590 patients were enrolled and randomly divided into a training group ($n = 456$) and a testing group ($n = 134$). The training group consisted of 268 male and 188 female patients, aged 43.4 (12–86) years old. The test group consisted of 67 male and 67 female patients aged 48.0 (12–79) years. A total of 1552 lumbar disc herniation lesion areas were marked in the sagittal and transverse MR images of the lumbar spine, with 1157 and 395 marked lesion areas in the training and testing groups, respectively. The lumbar spine MR images were annotated by a radiologist and then verified by a spinal surgeon.

4.2. Qualitative diagnostic identification results

By training on MR image data of 456 patients with lumbar disc herniation and testing with data from 134 patients, the artificial

intelligence model's FROC curve and PR curve for detecting lumbar disc herniation are shown in Fig. 2. The AUFROC and F1 values below the FROC curve in the sagittal MR image are 0.968 (95% CI: 0.943–0.994) and 0.971 (95% CI: 0.951–0.987), respectively. The AUFROC and F1 values in the transverse MR images were 0.893 (95% CI: 0.870–0.932) and 0.903 (95% CI: 0.870–0.935), respectively, demonstrating good detection performance. The accuracy and recall of the artificial intelligence model for detecting lumbar disc herniation on sagittal and transverse MR images are detailed in Table 1.

4.3. Segmentation results

The segmentation results of lumbar disc herniation lesions in the test set show that the artificial intelligence model can accurately recognize the shape and contour of the herniated disc tissue in lumbar MR images, as shown in Fig. 3. The red border-box in the figure represents the lesion area manually marked by the expert. In contrast, the green area represents the artificial intelligence model's automatic segmentation and recognition mark. We can see that artificial intelligence methods automatically segment and recognize marked areas, which are close to the lesion areas manually marked by experts, proving the importance of differential diagnosis for clinical lumbar disc herniation. Moreover, statistical analysis shows that based on each magnetic resonance image, the Dice value of lesion segmentation in the MR sagittal section image of the model is 0.774 ± 0.193 , and the Dice value of lesion segmentation in the MR transverse section image is 0.647 ± 0.261 , as shown in Fig. 3 (a); On a patient by patient basis, the dice value of lesion segmentation in the MR sagittal section image of the model is 0.777 ± 0.148 , and the Dice value of lesion segmentation in the MR transverse section image is 0.655 ± 0.180 , as shown in Fig. 3 (b). Whether at the 2D or 3D level, the segmentation accuracy of the model on MR sagittal section images is significantly higher than that on MR transverse section images ($P < 0.001$).

5. Discussion

Lower back pain is one of the most important causes of lifelong disability worldwide, leading to significant medical and social costs. Up to 97% of patients with low back pain are caused by degenerative

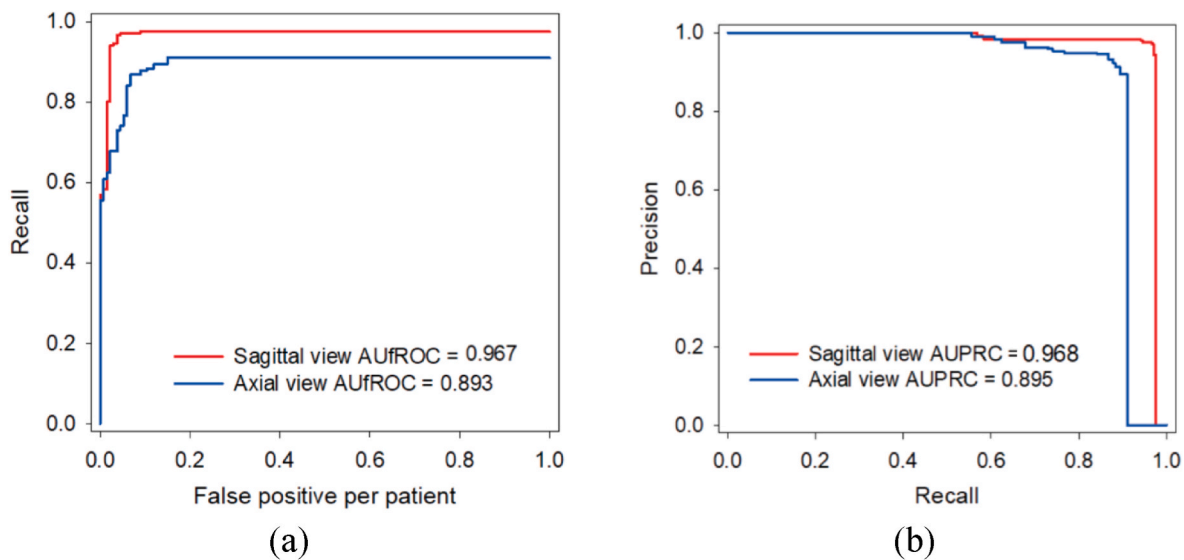


Fig. 2. FROC and PR curves on sagittal and transverse MR images in the test group.

Table 1
Accuracy of deep learning artificial intelligence in identifying lumbar disc herniation.

Recognition skills	Sagittal MR	Coronal MR
Precision (95% CI)	0.970 (0.943–0.995)	0.950 (0.943–0.995)
Recall (95% CI)	0.973 (0.947–0.990)	0.864 (0.818–0.909)
F1	0.972 (0.951–0.987)	0.902 (0.870–0.935)

changes in the lumbar spine (Yu et al., 2023). Degenerative changes in the lumbar spine occur in various anatomical structures, including intervertebral discs, endplates, sub-endplate bone, facet joints, ligaments, muscles, and soft tissues. Although 85% of cases have no precise diagnosis (Qiu et al., 2021b; Yu et al., 2023), studies have shown that modic changes in the vertebral body and intervertebral disc degeneration are essential factors (A et al., 2022; Gil et al., 2022). Lumbar magnetic resonance imaging can display abnormal changes in various components of the lumbar spine, providing a basis for the diagnosis and differential diagnosis of low back pain. Through imaging of soft tissues such as nerves and muscles, Lumbar MRI can display the degeneration of lumbar intervertebral discs, the degree and direction of disc herniation, and the compression of nerves in the spinal canal. At the same time, it can exclude tumor lesions in the spinal canal, making it the most accurate imaging method for diagnosing lumbar lesions to date, especially with high diagnostic value for lumbar disc herniation. In addition, MR examination can be used for the diagnosis and differential diagnosis of spinal malignant tumors, infections, trauma, and inflammatory diseases (Qiu et al., 2021c).

Good results have been achieved in the segmentation of lumbar disc herniation lesions, which is mainly related to the algorithm design of this study, in addition to using a large amount of data for training. Firstly, the Unet architecture model was adopted for lesion segmentation, considering the multi-scale and morphological differences in the lumbar disc herniation lesion area. The advantages of the pyramid structure were fully utilized, and features at different levels, including edges, shape, and semantics, were extracted through encoder down-sampling multiple times. The interactive fusion of multi-level features was achieved through a decoder (Yuan et al., 2023). Secondly, the correlation between sagittal and transverse MR images of patients was fully utilized. Using a multimodal mutual attention mechanism based on a transformer (Qiu et al., 2023), correlation analysis was performed on sagittal and transverse MR images, enhancing each feature through multimodal information, thereby achieving better segmentation results

and maximizing the utilization of multimodal information in the network. At the same time, the algorithm used also has certain limitations, mainly because U-net codecs are all local operations, and there is a certain amount of spatial information loss. In the future, we can consider adding a global aggregation module to aggregate global information and obtain more accurate segmentation results.

Spinal-related diseases are common in modern society, and the demand for MRI in clinical work is increasing. With the application of artificial intelligence technology in medical imaging work, it is possible to use software to automatically recognize the anatomical structure of the spine, detect lesions, and generate structured reports. It can alleviate the work pressure of radiologists and further explore the responsible lesions for clinical symptoms caused by lumbar MRI through deep learning methods, thereby improving the efficiency of clinical diagnosis and treatment.

6. Conclusion

The core contradiction in China's medical system is that the supply of medical resources with physicians as the core cannot meet the rapidly growing demand for patients, leading to a massive demand for medical artificial intelligence in China. In the process of disease diagnosis and treatment, low patient follow-up tracking rate, unstructured case data, busy clinical work, and high pressure are the difficulties in the work of orthopedic physicians. It is estimated that the outpatient diagnostic error rate in the United States is 5.08%, with 6 million patients suffering secondary harm each year due to incorrect diagnoses. The application of deep learning artificial intelligence is still in its early stages. It faces many challenges, such as incomplete data links, small data volumes, and unstable data on various platforms. Clinical big data is the cornerstone of AI development, and it is crucial to use unstructured big datasets to make them more intuitive and visible, ultimately transforming them into universal tools for healthcare professionals and patients.

This study demonstrates that an artificial intelligence recognition system based on deep learning can effectively analyze MR images of lumbar disc herniation, accurately identify the location and shape contour of the lesion, further improve the efficiency of clinical experts in reviewing films, and reduce misdiagnosis and missed diagnosis of diseases.

CRedit authorship contribution statement

Jun Qian: Writing – review & editing, Writing – original draft,

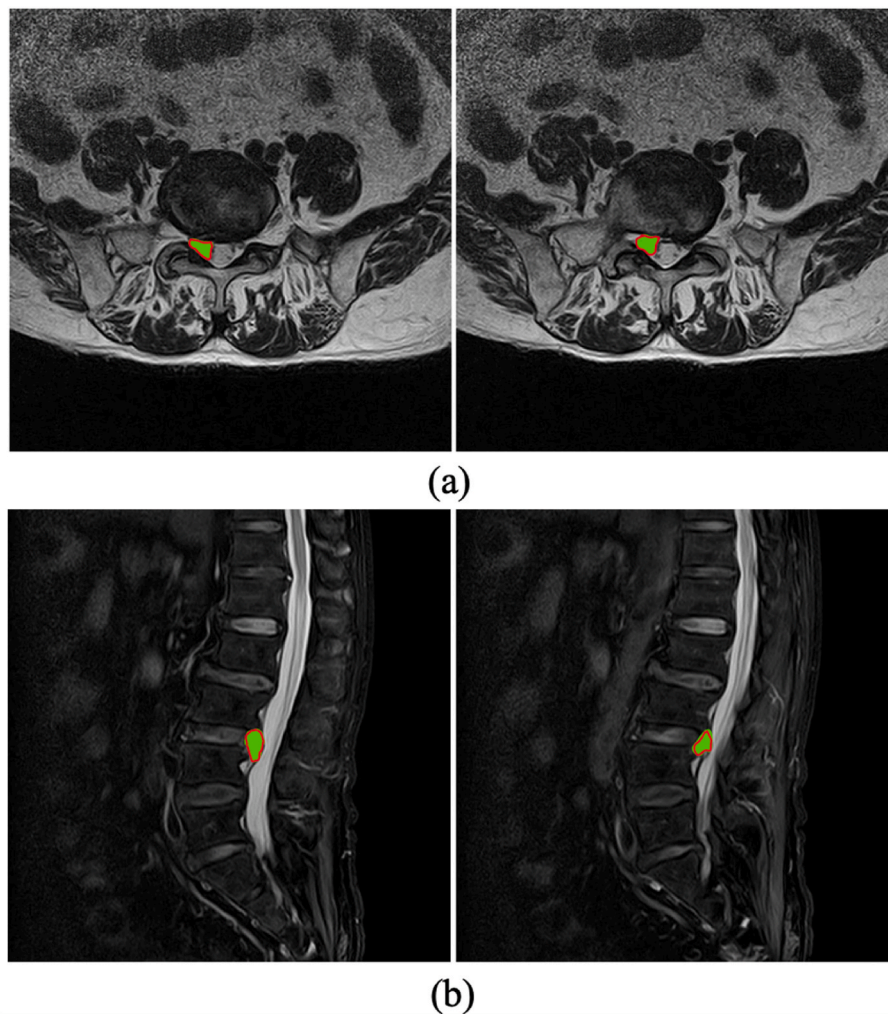


Fig. 3. Automatic segmentation results of MR images in patients with lumbar disc herniation. Red border box represents the lesion area manually marked by the expert, while the green area represents the automatic segmentation and recognition mark of the artificial intelligence model. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Methodology, Funding acquisition. **Guanghui Su:** Formal analysis, Data curation, Conceptualization. **Xiaolin Shu:** Supervision, Software, Resources. **Kuanghong Shen:** Visualization, Validation, Methodology, Investigation. **Bing Chen:** Visualization, Methodology, Investigation, Formal analysis. **Xiangdong Wang:** Writing – review & editing, Project administration, Methodology, Investigation.

References

- A, A. T. R., B, R. D. A., & C, S. G. D. S. (2022). Clinical and functional outcomes in patients submitted to early versus late surgery for lumbar disc herniation: A systematic review. *Interdisciplinary Neurosurgery*.
- Batcik, O. E., Kanat, A., Ozdemir, B., et al. (2021). Asymmetric CSA size of the psoas muscle is a critical feature in patients with L4-5 lumbar disc herniations. *Turkish Neurosurgery*. <https://doi.org/10.5137/1019-5149.JTN.34649-21.2>
- Bhattarai, U., Gurung, P., Shrestha, J., et al. (2023). A case of intradural lumbar disc herniation. *Clinical Case Reports*, 11(6). <https://doi.org/10.1002/ccr3.7514>. n/a-n/a.
- Cheung, K. T., & Hong, H. (2023). Effect of heat therapy using sea salt combined with Chinese herbs on chronic low-back pain due to lumbar disc herniation: A randomized, single-blinded, controlled feasibility study. *Medical Acupuncture*, (1), 35.
- Gadjradj, P., Harhangi, B. S., Amelink, J., et al. (2020). Percutaneous transforaminal endoscopic discectomy versus open microdiscectomy for lumbar disc herniation: A systematic review and meta-analysis. *Spine*. <https://doi.org/10.1097/BRS.0000000000003843>
- Gil, S.-L., Eisenberg, E., Zohar, Y., et al. (2022). Mechanism underlying painful radiculopathy in patients with lumbar disc herniation. *European Journal of Pain*, (6), 26.
- He, Q., Zhao, J., Fan, M., et al. (2020). Effect of continuous nursing based on WeChat platform on postoperative rehabilitation of patients with lumbar disc herniation. *Japan Journal of Nursing Science*. , Article e12382. <https://doi.org/10.1111/jjns.12382>
- Mo, Z., Li, D., Zhang, R., et al. (2019). Comparisons of the effectiveness and safety of tuina, acupuncture, traction, and Chinese herbs for lumbar disc herniation: A systematic review and network meta-analysis. *Evidence-based Complementary and Alternative Medicine*, 2019, 1–10. <https://doi.org/10.1155/2019/6821310>
- Qiu, D., Cheng, Y., & Wang, X. (2021b). Gradual back-projection residual attention network for magnetic resonance image super-resolution. *Computer Methods and Programs in Biomedicine*, 208, Article 106252.
- Qiu, D., Cheng, Y., & Wang, X. (2021c). Progressive U-net residual network for computed tomography images super-resolution in the screening of COVID-19. *Journal of Radiation Research and Applied Sciences*, 14(1), 369–379.
- Qiu, D., Cheng, Y., & Wang, X. (2023). Medical image super-resolution reconstruction algorithms based on deep learning: A survey. *Computer Methods and Programs in Biomedicine*, 238, Article 107590.
- Qiu, D., Zheng, L., Zhu, J., et al. (2021a). Multiple improved residual networks for medical image super-resolution. *Future Generation Computer Systems*, 200–208.
- Shi, Y. Z., FengDou, W. D., HongyuanZou, Y. F., YangWang, C. S., & HaibinRen, Y. (2021). Quantitative evaluation of intraspinal lumbar disc herniation-related lumbosacral radiculopathy before and after percutaneous transforaminal endoscopic discectomy using diffusion tensor imaging. *Spine*, 46(13).
- Soo, E. S. (2020). Can body mass index affect the size of lumbar disc herniation? Results from an age and gender-matched case-control study. *Neurosurgery*, (Supplement_1) https://doi.org/10.1093/neuros/nyaa447_759. Supplement_1.
- Wang, L., Zhou, C., Liu, X., et al. (2021). Study on influencing factors of postoperative recurrence of lumbar disc herniation treated by percutaneous transforaminal endoscopic discectomy. *Indian Journal of Pharmaceutical Sciences*, (6), 83.
- Wu, X., Ma, Y., Ding, R., et al. (2021). Should adjacent asymptomatic lumbar disc herniation be simultaneously rectified? A retrospective cohort study of 371 cases

- that only received an open fusion or endoscopic discectomy on symptomatic segments. *The Spine Journal*, 21(3).
- Wu, B., Zhan, G., Tian, X., et al. (2019). Comparison of transforaminal percutaneous endoscopic lumbar discectomy with and without foraminoplasty for lumbar disc herniation: A 2-year follow-up. *Pain Research and Management*, 2019, 1–12. <https://doi.org/10.1155/2019/6924941>
- Yu, P., Qiang, H., Zhou, J., et al. (2019). Percutaneous transforaminal endoscopic discectomy versus micro-endoscopic discectomy for lumbar disc herniation. *Medical Science Monitor*. <https://doi.org/10.12659/msm.913326>
- Yu, X., Yue, H., Wei, H., et al. (2023). Comparative study of unilateral biportal endoscopic and traditional open surgery in treating lumbar disc herniation. *Alternative Therapies in Health & Medicine*.
- Yuan, P., Shi, X., Wei, X., et al. (2022). Development process and clinical application of collagenase chemonucleolysis in treating lumbar disc herniation: A narrative review in China. *Postgraduate Medical Journal*. <https://doi.org/10.1136/postgradmedj-2021-141208>
- Yuan, P., Shi, X., Wei, X., et al. (2023). Development process and clinical application of collagenase chemonucleolysis in treating lumbar disc herniation: A narrative review in China. *Postgraduate Medical Journal*, (1172), 99.
- Zhou, S. (2022). A comparative study on treating lumbar disc herniation by traditional Chinese medical massage. *Traditional Medicine and Modern Medicine*, 5(01n04), 13–17. <https://doi.org/10.1142/S2575900023300023>